

SC7 Bayes Methods MT25

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Lecture 16: A Gibbs sampler for partitions of the Galaxy data

Gibbs sampler for the partitions

- Let $X_t = (\theta^*, S)$ with $S = (S_1, \dots, S_K)$ and $\theta^* = (\theta_1^*, \dots, \theta_K^*)$ be the current state of the chain (with $n_k = |S_k|$, $k = 1, \dots, K$)
- We need an update that operates on S (and θ^*). This must be doing reversible jump as $\dim(\theta^*)$ changes if K changes.
- pick an entry $i \in [n]$ at random, take it out of its cluster, k_i say, and move it to a randomly chosen cluster.
- could be cluster $K + 1$: add a cluster and increase dimension.
- removing i from cluster k_i may empty that cluster: remove that cluster and decrease dimension.
- following proposal has acceptance probability equal one!*

*Algorithm 8 in RM Neal, "Markov Chain Sampling Methods for Dirichlet Process Mixture Models", JCGS 9:249-265 (2000), but different proof.

Intuition

The generative model for θ^*, S, y from $K = 0$ and $S = \emptyset$ is

1. For $j = 0, \dots, n - 1$,

(a) Set $q(S) = (q_1(S), \dots, q_{K+1}(S))$ with

$$q_k(S) = \begin{cases} \alpha & \text{if } k = K + 1, \\ n_k = |S_k| & \text{for } k \in [K], \end{cases}$$

set $p(S) = q(S) / \sum_k q_k(S)$ and simulate $k \sim p(S)$.

(b) if $k \in [K]$, set $S_k \leftarrow S_k \cup \{j + 1\}$ and $y_{j+1} \sim f(\cdot | \theta_k^*)$.

(c) if $k = K + 1$, set $S_{K+1} = \{j + 1\}$ and $K \leftarrow K + 1$;
simulate $\theta_{K+1}^* \sim H$ and $y_{j+1} \sim f(\cdot | \theta_{K+1}^*)$.

If $\theta_j = \theta(\theta^*, S)_j$ then (θ_j, y_j) 's are exchangeable so make i the last arrival and get the same generative model.

Remove i from its cluster; we are at step $n - 1$ of the CRP; put i into a random cluster for new partition $S \rightarrow S^{-i} \rightarrow S'$.

Simulation conditional on y (Case I, $|S_{k_i}| > 1$)

If $|S_{k_i}| > 1$ then $S \rightarrow S^{-i} = (S_1^{-i}, \dots, S_K^{-i}) \rightarrow S'$.

In the posterior we condition on known y_i so if we put i in S'_k then we are saying y_i was drawn as $y_i \sim f(\cdot | \theta_k^*)$.

The choice of $k \in \{1, \dots, K+1\}$ now weighted by $f(\cdot | \theta_{k'}^*)$ so

$$\vec{q}_k(S, k_i) = \begin{cases} (n_{k_i} - 1)f(y_i | \theta_{k_i}^*) & \text{if } k = k_i, \\ \alpha f(y_i | \theta'_{K+1}) & \text{if } k = K+1, \\ n_k f(y_i | \theta_k^*) & \text{for } k \in [K] \setminus \{k_i\}. \end{cases}$$

for $k \in [K+1]$.

Notice that if we choose $k = K+1$ it's a new table so we need a new $\theta_{K+1}^* \sim h$ before we assign i to a cluster.

Set $\vec{q}(S, k_i) = (\vec{q}_1(S, k_i), \dots, \vec{q}_{K+1}(S, k_i))$,

$$\vec{p}(S, k_i) = \vec{q}(S, k_i) / \sum_k \vec{q}_k(S, k_i)$$

simulate $k \sim \vec{p}(S, k_i)$ and put i in S'_k . Candidate S' is

$$S' \leftarrow S, S'_{k_i} \leftarrow S'_{k_i} \setminus \{i\} \text{ and } S'_k \leftarrow S'_k \cup \{i\}.$$

Simulation conditional on y (Case II, $|S_{k_i}| = 1$).

When i arrived we had K^{-i} clusters. Index S_k by arrival order.

If $|S_{k_i}| = 1$ then S had $S_K = \{i\}$ so $k_i = K$ and when i arrived the partition was $S^{-i} = (S_1, \dots, S_{K-i})$ with $K^{-i} = K - 1$.

The choice of $k \in \{1, \dots, K^{-i} + 1\}$ is now weighted by

$$\tilde{q}_k(S, k_i) = \begin{cases} \alpha f(y_i | \theta_{k_i}^*) & \text{if } k = k_i, \\ n_k f(y_i | \theta_k^*) & \text{for } k \in [K] \setminus \{k_i\}. \end{cases}$$

for $k \in [K]$ (don't insist $k_i = K$ as this is old indexing in S).

The choice $k = k_i$ is a new table but dish $\theta_{k_i}^*$ is already served.

Set $\tilde{q}(S, k_i) = (\tilde{q}_1(S, k_i), \dots, \tilde{q}_K(S, k_i))$,

$$\tilde{p}(S, k_i) = \tilde{q}(S, k_i) / \sum_k \tilde{q}_k(S, k_i)$$

simulate $k \sim \tilde{p}(S, k_i)$ and put i in S_k . Candidate S' is

$$S' \leftarrow S, S'_{k_i} \leftarrow S'_{k_i} \setminus \{i\} \text{ and } S'_k \leftarrow S'_k \cup \{i\}.$$

If $k \neq k_i$ then $S'_{k_i} = \emptyset$ is empty so remove it and re-index sets.

Gibbs sampler targeting $\pi(\theta^*, S|y)$

Suppose $X_t = (\theta^*, S)$. Proposal will be (θ', S') .

1. Choose $i \sim U\{1, \dots, n\}$ and suppose $i \in S_{k_i}$.
2. If $n_{k_i} > 1$ - removing i wont empty S_{k_i} .
Simulate $\theta'_{K+1} \sim h(\cdot)$ and proceed as Case I to get S' .
 - (a) If $K' = K$ then set $\theta' = \theta^*$
 - (b) Else (if $K' = K + 1$) set $\theta' \leftarrow \theta' \cup \{\theta'_{K+1}\}$.
3. Else if $n_{k_i} = 1$ - removing i will empty S_{k_i} .
Proceed as Case II. This gives S' .
 - (a) If $K' = K$ then set $\theta' = \theta^*$
 - (b) Else (if $K' = K - 1$) set $\theta' = \theta^* \setminus \{\theta^*_{k_i}\}$.
4. $X_{t+1} = (\theta', S')$.

Dimension up in 2(b), down in 3(b), unchanged in 2(a) and 3(a).

The acceptance probabilities for these proposals all equal one.

Proof acceptance probability equals one

Up in 2(b): from (θ', S') with $n_{k_i} > 1$; choose $k = K + 1$ using $k \sim \vec{p}(S, k_i)$ and simulate $\theta'_{K+1} \sim h$ to get $\theta' = (\theta^*, \theta'_{K+1})$.

Back in 3(b): from (θ', S') , and we chose the same i and $k = k_i$ using $k \sim \tilde{p}(S', K + 1)$.

$$\begin{aligned}
 \alpha(\theta', S' | \theta^*, S) &= \min \left\{ 1, \frac{\pi(\theta', S' | y)}{\pi(\theta^*, S | y)} \times \frac{n^{-1} \tilde{p}_{k_i}(S', K + 1)}{n^{-1} h(\theta'_{K+1}) \vec{p}_{K+1}(S, k_i)} \right\} \\
 &= \min \left\{ 1, \frac{f(y_i | \theta'_{K+1}) h(\theta'_{K+1}) \alpha^{K+1} \Gamma(n_{K+1}) \Gamma(n_{k_i} - 1)}{f(y_i | \theta_{k_i}^*) \alpha^K \Gamma(n_{k_i})} \right. \\
 &\quad \left. \times \frac{\tilde{q}_{k_i}(S', K + 1)}{h(\theta'_{K+1}) \vec{q}_{K+1}(S, k_i)} \right\} \\
 &= \min \left\{ 1, \frac{f(y_i | \theta'_{K+1}) h(\theta'_{K+1}) \alpha}{f(y_i | \theta_{k_i}^*) (n_{k_i} - 1)} \times \frac{(n_{k_i} - 1) f(y_i | \theta_{k_i}^*)}{h(\theta'_{K+1}) \alpha f(y_i | \theta'_{K+1})} \right\} \\
 &= 1.
 \end{aligned}$$

Key is denominators in $\tilde{p}_{k_i}(S', K + 1)$ and $\vec{p}_{K+1}(S, k_i)$ cancel.

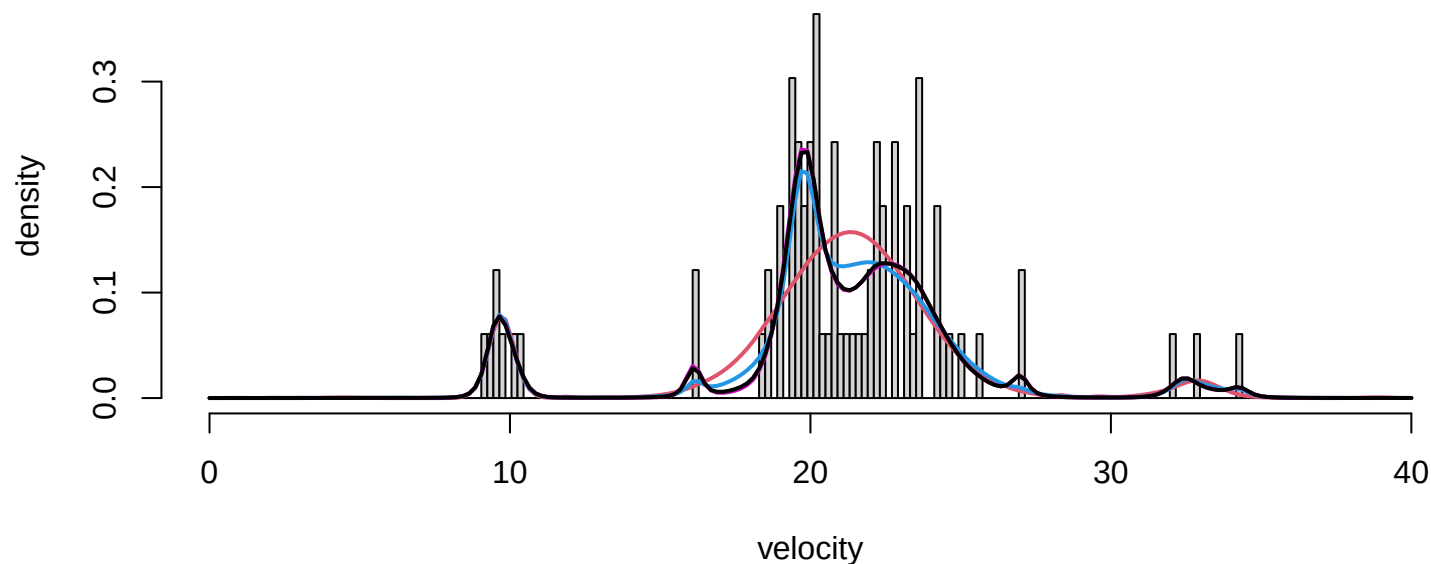
$$\begin{aligned}
 \sum_{j=1}^{K(S')} \tilde{q}_j(S', K + 1) &= n'_1 f(y_i | \theta'_1) + \dots + n'_{k_i} f(y_i | \theta'_{k_i}) + \dots + \alpha f(y_i | \theta'_{K(S')}) \\
 &= n_1 f(y_i | \theta'_1) + \dots + (n_{k_i} - 1) f(y_i | \theta'_{k_i}) + \dots + \alpha f(y_i | \theta'_{K(S)+1}) \\
 &= \sum_{j=1}^{K(S)+1} \vec{q}_j(S, k_i).
 \end{aligned}$$

For middle term in the top row is $\tilde{q}_{k_i}(S', K + 1)$. In the second row it is $\vec{q}_{k_i}(S', k_i)$. These equal as $n'_{k_i} = n_{k_i} - 1$.

The last term on the top line is $\tilde{q}_{K+1}(S', K + 1)$. This is the Case II setting where we put i back in the cluster we just emptied.

The last term on the second line is $\vec{q}_{K+1}(S, k_i)$. This is going forward and we want the weight to put i in a new cluster.

Each term has a matching pair so the sums are equal. For other cases see notes.



We implemented the two sets of updates and ran the MCMC. Output analysis is similar to RJ-MCMC case study.

We estimate the posterior predictive distribution (black) and the predictive distribution conditioned on 3,4 and 5 components (red, green and blue) and plot these PPD's over the data, as we did in lecture 13 for the RJ-MCMC fit. See associated R-code for this lecture for implementation and further output summaries.